

Judgment rules & Approve → Zoho

Document Intelligence Connector — quick reference for Phase steps **1.2** (judgment) and **1.3** (Zoho push on human approve).

1. Automatic judgment rules (only these three)

These run when you call the `judgment` function (also after upload / **Run extract + judgment** in the UI). Results are stored as rows in `judgment_results`.

Rule name	What it checks	Passes when	Fails when
<code>duplicate_vendor_amount_date</code>	Same vendor + amount + invoice date	No matching peer document in the same company within the day window	Same vendor + amount + date found within the window
<code>missing_supporting_document</code>	Supporting document present	<code>documents.has_supporting_document = true</code>	Flag is false / missing
<code>amount_above_threshold_no_po</code>	High amount without a PO	Amount $\leq$ company threshold or <code>po_number</code> is set	Amount above threshold and no PO

Config (per company — `company_config`)

- `duplicate_check_days` — default **3**
- `amount_requires_po_threshold` — default **5000** (may be changed for tests, e.g. 1000)

Not a fourth automatic rule

On **Approve**, the UI may insert `human_review_approval` or mark failed rules as passed with “Approved by {reviewer}”. That is a **human override / audit** row, not part of the automatic 3-check set.

The pluggable rules engine scaffold (`judgment/engine.ts`) has **no registered rules** yet — only the three hardcoded checks run.

2. What Approve → Zoho does

Preconditions

- Document must have an `extracted_fields` row (vendor, amount, date).
- Button is disabled if fields are missing or status is already `synced`.
- Edge functions must be running:
`npx supabase functions serve --env-file .env`

Step-by-step

- Human override** — any failed judgment rows are set to `passed: true` (noted as approved by reviewer).
If none failed, insert `human_review_approval`.
- Status** → `approved`.

3. `zoho-push` runs:

- Confirms human approval + all judgment rows pass
- Resolves or **auto-creates** vendor in the Zoho org from `.env`
- Uses expense account from `ZOHO_DEFAULT_ACCOUNT_ID` / `ZOHO_EXPENSE_CATEGORY`
- Creates a **Bill** (with explicit `bill_number`)
- **Attaches** the invoice file from Storage
- Writes `erp_sync_log` with Zoho `bill_id`

4. **Status** → `synced` on success.

Outcomes

Result	Document status	Meaning
Success	<code>synced</code>	Bill created in Zoho; attachment uploaded; <code>erp_sync_log</code> has <code>external_doc_id</code>
Push failed	<code>approved</code>	App approved, but Zoho did not complete — check the green/red message in the UI
Blocked	<code>unchanged</code>	No extract yet — Approve refused

Approve is **not** a dry run: success creates a real bill in your Zoho test organization.

3. How to know something was pushed

1. **UI** — status `synced` , “Zoho bill” id shown, success message with bill id + attachment note
2. **Studio** — table `erp_sync_log` → `external_doc_id` = Zoho bill id
3. **Zoho Books** — switch to org `ZOHO_ORGANIZATION_ID` → Bills → open that bill → confirm PDF attachment

4. Suggested test flow in the UI

1. Start Supabase + functions serve + web UI (`npm run web:dev`).
2. Upload a clear invoice (or **Run extract + judgment** on an existing upload).
3. Confirm the right panel shows **this document’s** vendor / amount / date (not empty).
4. Review the three judgment rows (pass/fail).
5. Click **Approve** → **Zoho**.
6. Verify `synced` + Zoho bill + attachment.

Example expectations

- **Clear invoice with PO** (e.g. Bright Peak + PO number) — amount/PO check usually **passes**; good Approve candidate.
- **High amount, no PO** — amount/PO check usually **fails** until you Correct a PO or Approve overrides failed rules.

5. Related files

Area	Path
Checks	<code>supabase/functions/judgment/checks.ts</code>
Judgment runner	<code>supabase/functions/judgment/index.ts</code>

Zoho push	supabase/functions/zoho-push/index.ts
Review UI Approve	apps/web/src/components/ReviewPanel.tsx
Rules registry	docs/rules-registry.md